Lumpy Skin Disease Virus Antibody High-Sensitivity Fluorescence Detection Kit

Instruction manual

For Veterinary Use Only

【Product Name】 The Gaomin Fluorescence Detection KIT of Antibody to Lumpy Skin Disease Virus

【Kit Components】

Components	数量	装量
Immunoassay Strip	20 Pouches/Box、50 Pouches/Box	1 Unit/Pouch
Sample Dilute	1 Bottle/Box	12 mL/Bottle or 30 mL/Bottle
EP Pipette	1 Packet/Box	25 pcs or 55 pcs
Instructions for Use (IFU)	1 Piece/Box	1 Piece

【Indications and Usage】For rapid quantitative detection of antibodies or immune titers against bovine nodular skin disease virus in bovine serum or plasma, either on-site or in the laboratory.

【Significance of Testing】 Lumpy skin disease (LSD), also known as bovine nodular skin disease, is an acute, subacute, or chronic infectious disease in cattle caused by the lumpy skin disease virus. Clinically, it is mainly characterized by fever, emaciation, swollen lymph nodes, skin edema, and the formation of hard nodules or ulcers in specific areas. Infected cattle show reduced milk production, and the hides may have depressions or holes after tanning, greatly reducing their economic value. Rapid and convenient testing of antibodies against the lumpy skin disease virus allows for an objective evaluation of vaccine efficacy or determination of infection status, providing a reliable basis for developing treatment and prevention measures.

【Compatible Instruments】 WellRay Fluorescence Immunoassay Analyzers (Manufacturer: Chengdu WellRay Biotechnology Co., Ltd.), Supported Models: WR-1608D/WR-1908/WR-1860/DR-1111.

【Sample Requirements】 Collect bovine serum and plasma using standard procedures, use immediately, and store in a refrigerator at 2-8°C for no more than 5 days.

【Procedures and Interpretation】

1 Procedures

1.1 Preparation

(1) Allow the test cassette, sample working solution, and test sample to reach room temperature.

Note: Ensure that all components are equilibrated to 20-35°C. The test cassette must be used within 5 minutes after opening the foil pouch.

(2) Power on: Press and hold the power button (until the indicator light turns on) on the high-sensitivity fluorescence analyzer.

(3) Download and install the software by scanning the QR code at the end of the manual with your smartphone.

(4) Open the software and scan the QR code on the test kit with your smartphone. For computer-based detection, manually enter the test kit identification code.

(5) Select the identification code item for this batch of test kits.

(6) Label the test cassette with the corresponding sample name or code (Recommended: Sequentially number them according to the serial number and arrange them in order).

1.2 High-Sensitivity Fluorescence Analyzer Technical Specifications

1.2.1 Basic Parameters

(1) Unit Dimensions: 150×95×55 (mm)

(2) Net Weight: ≤ 0.6 kg

(3) Excitation Wavelength: 400 nm±5 nm

(4) Emission Wavelength: 615 nm±3 nm

1.2.2 Operating Conditions

(1) Power Source: Charger or USB

(2) Power Specifications：DC 5V-1A

(3) Ambient Temperature：5℃-40℃

(4) Relative Humidity：≤80%

(5) Avoid high-intensity electromagnetic fields.

(6) Keep away from direct strong light

1.3 Dilution: Dilute the test sample 50 times: first, take 490 μL of working solution and add it to an EP tube, then add 10 μL of the bovine serum or blood sample to be tested.

1.4 Sample Addition

(1) Set the pipette to 80μL and repeatedly aspirate and dispense 5 times, or use a vortex mixer to mix the sample to be tested;

(2) Take 80μL of the mixed sample and add it to the sample well of the reagent card corresponding to the sample number. Let the immune reaction stand for 15 minutes before testing.

1.5 Detection

Insert the test cassette that has reacted for 15 minutes into the high-sensitivity fluorescence analyzer, fill in the corresponding sample ID, then click "Detect" to instantly display the test results on your mobile phone or computer.

1.6 Batch Testing

For large sample volumes, sequential sample addition and incubation intervals can be designed to maintain relatively consistent timing, enabling continuous high-throughput testing (200 samples/hour). By selecting the "Incremental" detection mode, results are automatically and systematically generated, eliminating the need for manually entering each sample name.

2 Interpretation

Result interruption：

Results < 16 are negative;

Results ≥ 16 are positive;

The measured antibody **Titer value** is proportional to the content of the bovine nodular skin disease virus.

【Precautions】

(1) Before testing, inspect the aluminum foil pouch for damage; if compromised, do not use to avoid affecting test results.

(2) Components and standard curves from different reagent lot numbers must not be mixed, as standard curves are traceably established through precise detection and calculation.

(3) The optimal detection environment temperature is 20°C to 35°C. The incubation process must be protected from strong light exposure.

(4) Use the test strip immediately after opening the aluminum foil pouch to prevent environmental humidity from affecting its performance.

(5) This kit is for screening purposes only, and the results are intended for reference in veterinary clinical diagnosis. If the test result is "Invalid", please retest.

(6) Vortex mixing or repeated pipetting is recommended prior to sample loading to ensure homogeneity and prevent

detection errors caused by uneven samples.

(7) Please read the high-sensitivity fluorescence analyzer manual carefully to master the operation method for "Detection".

(8) All test samples must be processed in strict accordance with regulations. Operators should wear disposable gloves and lab coats, and perform thorough site disinfection upon completion of testing.

【Specifications】(1) 20 Pouches/Box (2) 50 Pouches/Box

【Storage and Validity Period】Transport at ambient temperature, store at 2-8°C, with a shelf life of 12 months.

【Manufacturer】Hangzhou Darui Technology Co., Ltd.

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【Contact Information】Telephone number: 4000801860

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(Scan the QR code above to download the mobile testing software.)